



REPUBLIC OF YEMEN
SANA'A UNIVERSITY
FACULTY OF ENGINEERING
MECHATRONICS ENGINEERING
DEPARTMENT



Conversion from direct current (AC)
to alternating current(DC).

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202274198

Group No: The first group.

System(Parallel.)

Supervised by:

Mohammed AL_hitari.

Contact.

1_abstract.

2_Introduction.

3_procedure

A_Tolls

#_soft Ware.

#_Hard were.

B_Designs proces.

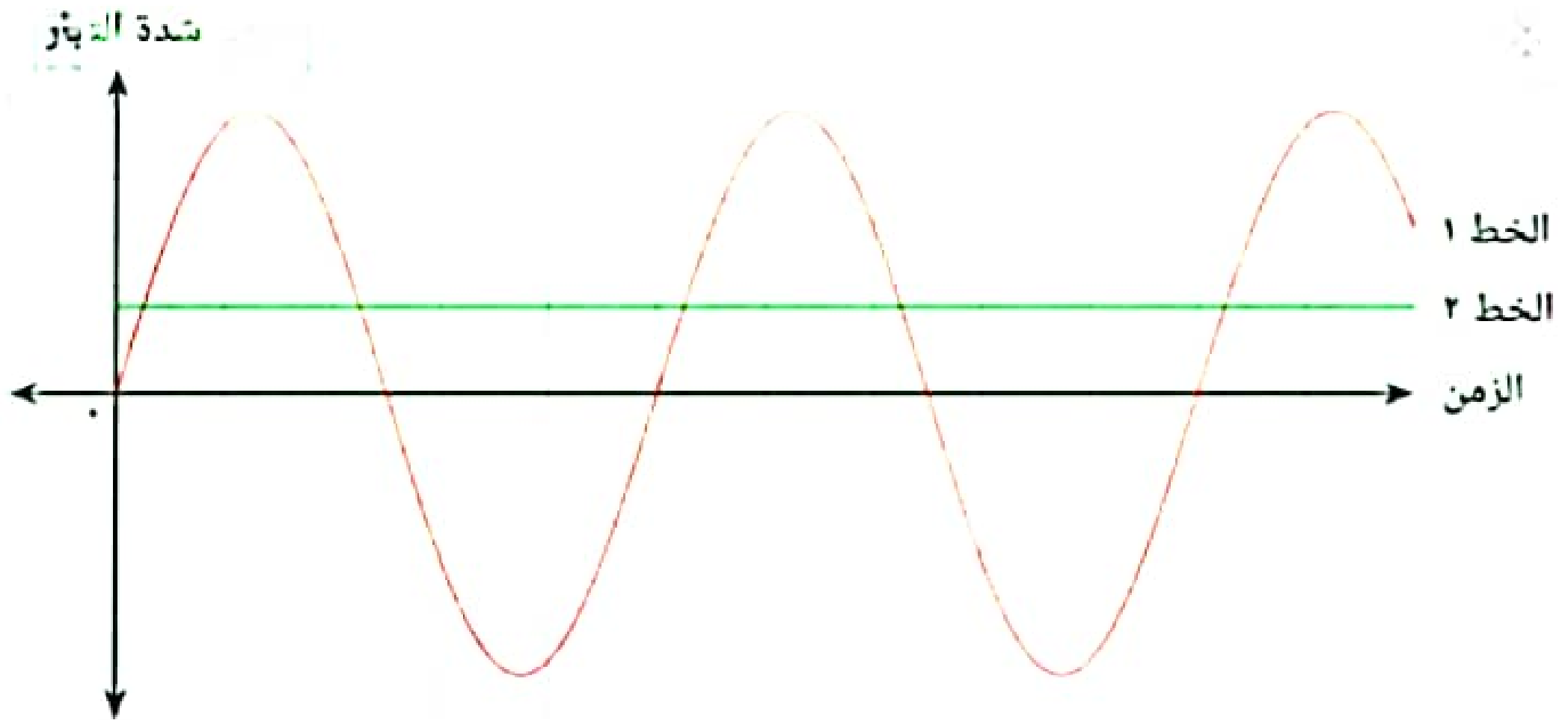
4_Descusion.

5_conclusion.

6_Refrence.

Introduction.

Alternating current is the most efficient way to deliver electrical power, but most electronic devices need DC to operate, which calls for the use of AC to DC converters as an integrated part of the devices themselves or as part of the power cable.



The form of the primary wave of AC before any proces.

Procedures.

Safety measures were taken through the work on several methods, where the wires were isolated from contact and the used pieces were installed well.

A_Tools

#_Software.

Hand-designed in the form of a sketch in the notebook . Also searched in Google for the shape of the waves before and after the conversion.

#_Hard were.

_Magnetic transducer.

_Diode bridge.

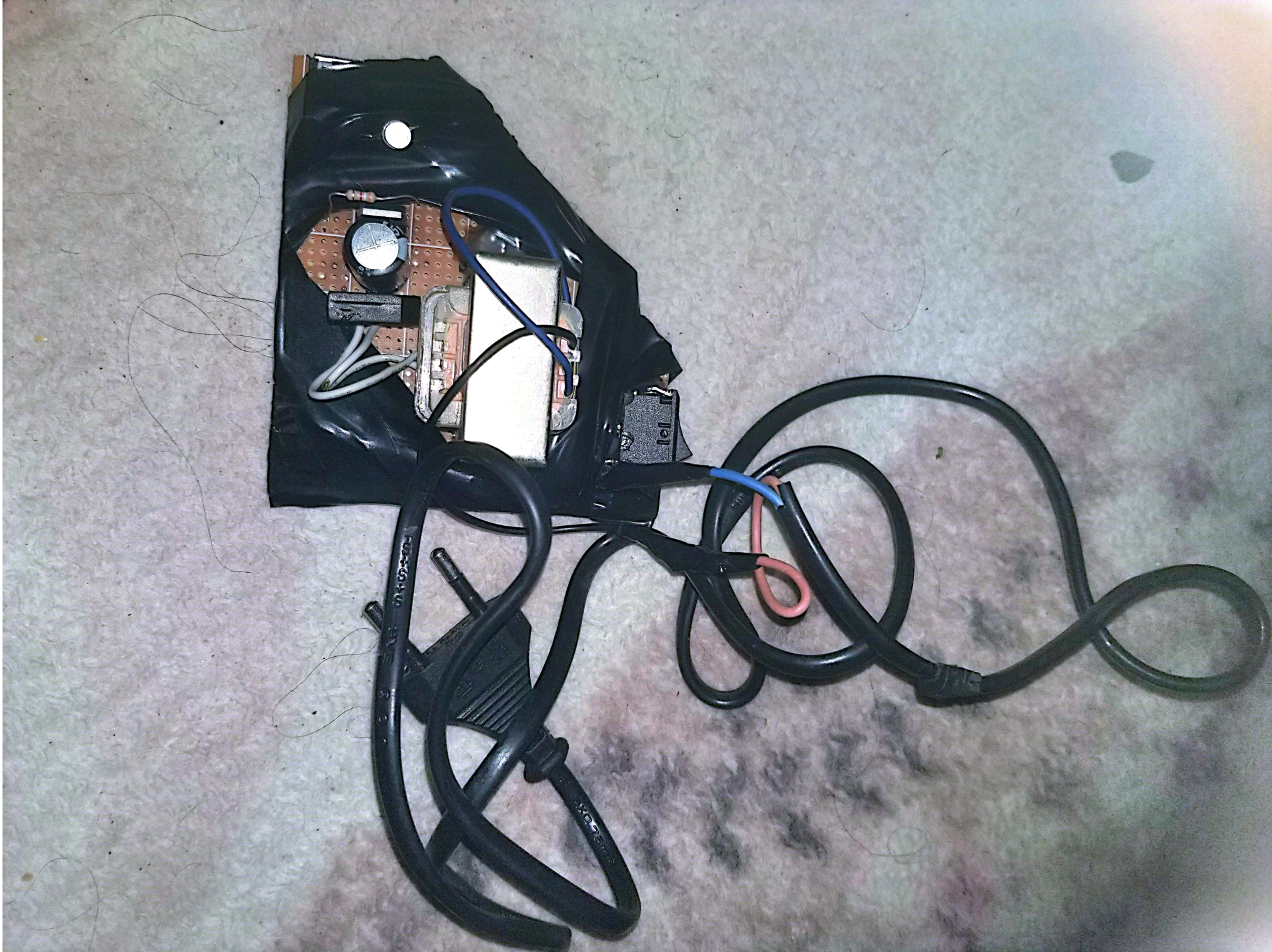
_Capacitor Capacity 2200.

_PNP . transistor.¹

_100 ohm resistance.

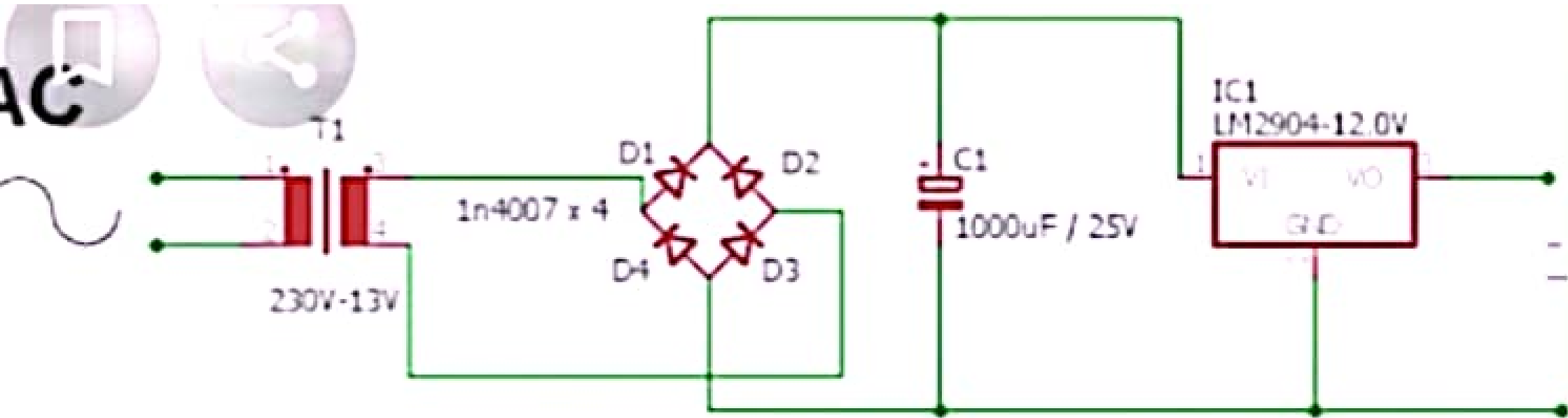
_LED.

_Swith.



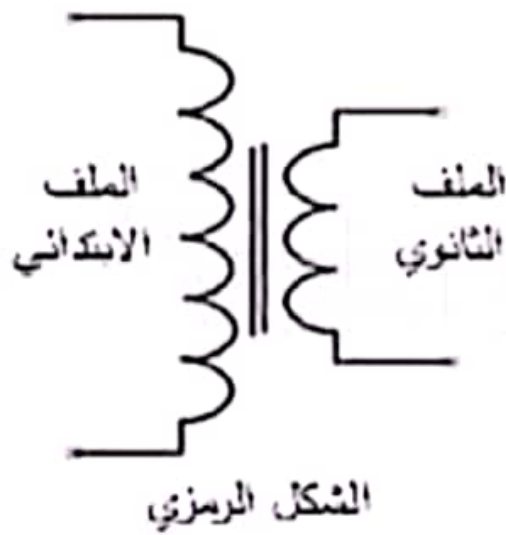
B_Designs proces.

The secondary coil with the greatest resistance was connected in the electric transformer to the alternating current, where it reduced the voltage to enter the alternating current directly into the primary coil with the lowest resistance. In this way, the voltage was reduced from 220 volts to 14 volts, and the transformer outputs the remaining voltage to the diode bridge and where the diode bridge By inserting alternating current and raising the value of the current, this changed the shape of the waves to a positive value, and since the wave shape has not changed yet to a constant current, the positive electrode of the capacitor was connected to the positive electrode of the diode and the negative electrode to negative, and this led to a change in the shape of the waves. Because of the capacitor, because the process of discharging with the capacitor is slow, and then the PNP transistor is connected where the pole P is connected to positive and the base called N is negative, and then the poles P and N connected to the negative and positive of the capacitor are connected with the bridge respectively to the negative pole of the output IC and the third pole of the transistor is connected P but The anode of the IC. To ensure that the current is constant, a 220 ohm resistor was connected to the positive pole of the transistor and the other direction to the positive pole of the LED, and the negative pole was connected to the negative pole.



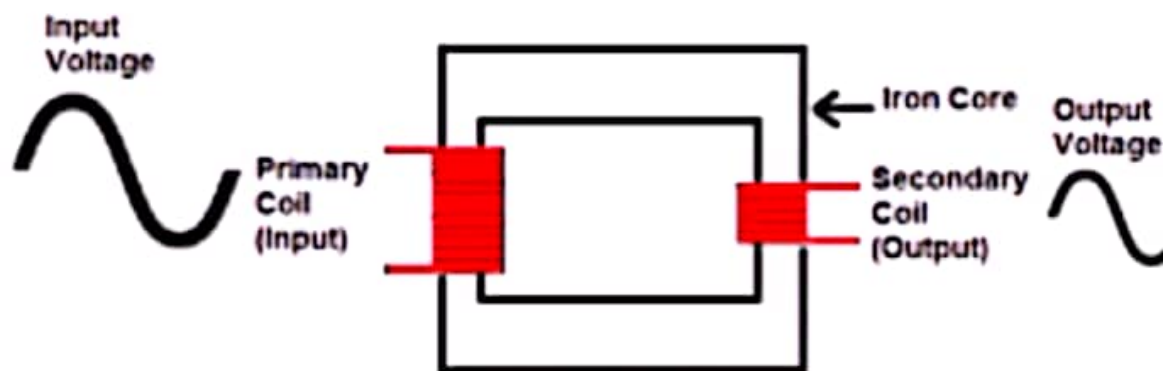
دائرة تحويل التيار من متناوب الى مستمر

The electrical circuit of the transformer from AC to DC.



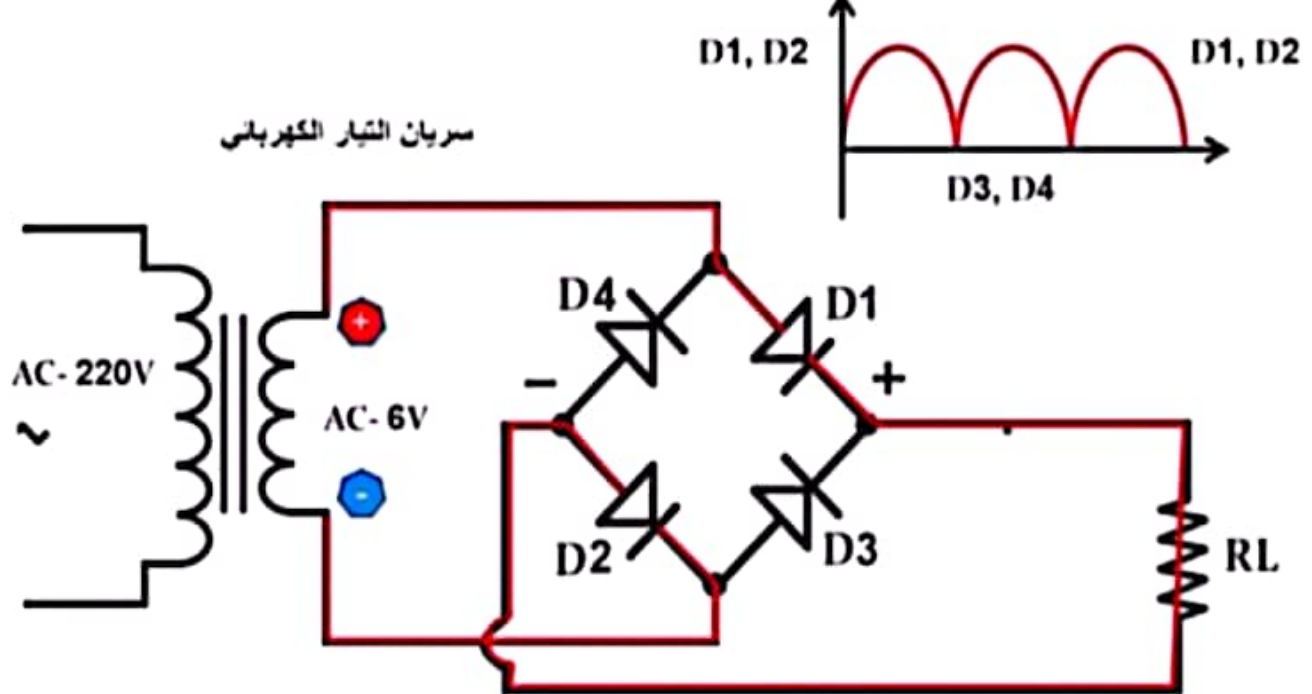
شكل المحوّل الخافض
(أربعة أطراف)

The transformer is a device that converts electrical power called volts, as it converts from 220 volts to approximately 14 volts.

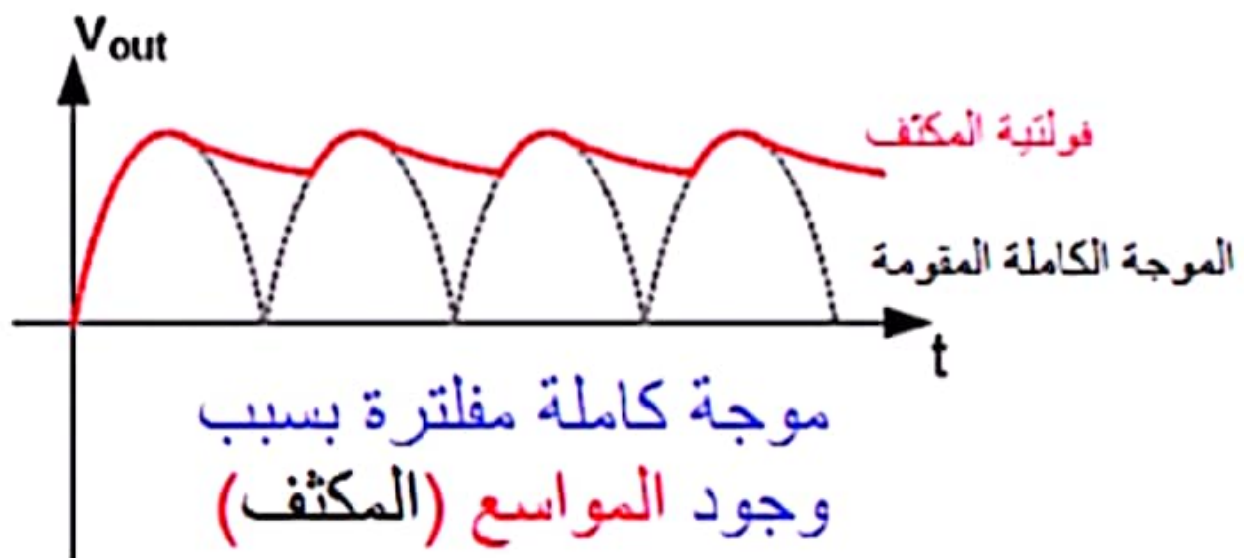


Step Down Transformer

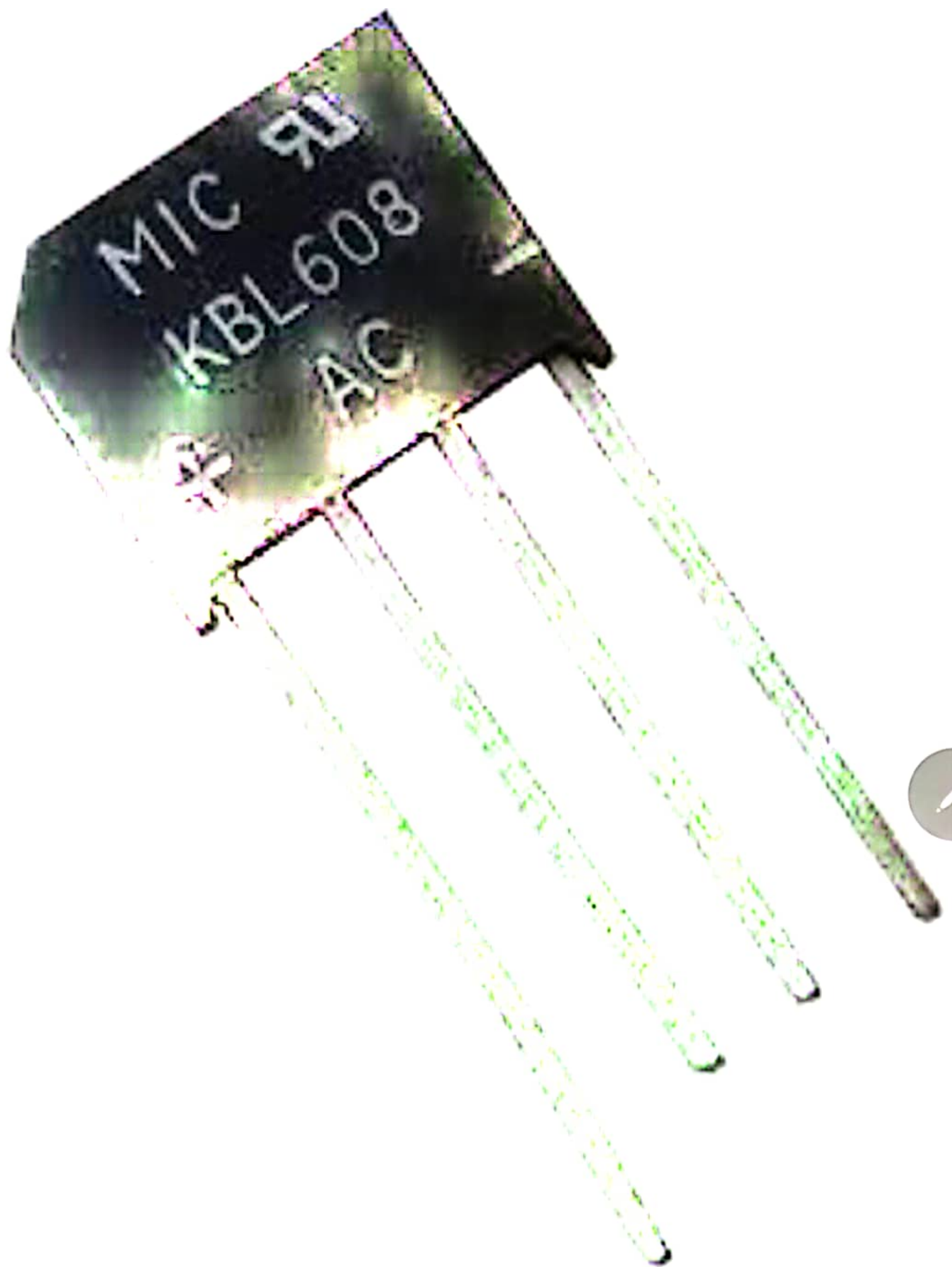
The shape of the waves after its exit from the converter.



The shape of the waves after passing through the diode bridge.

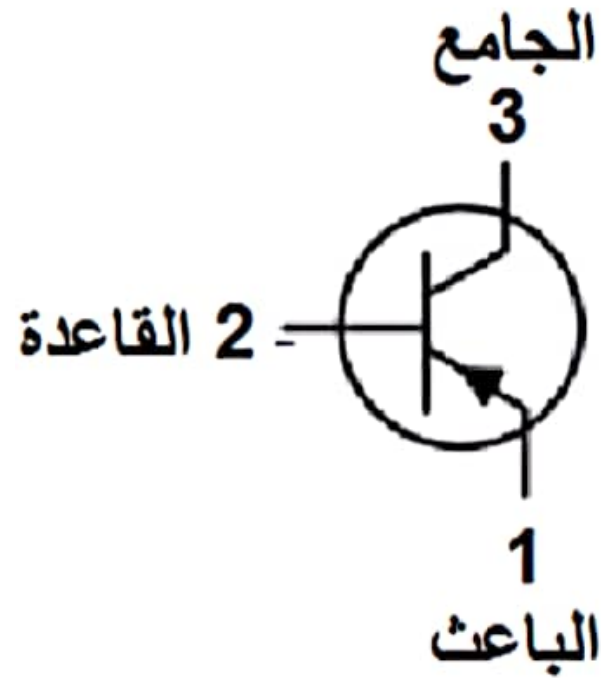


The shape of the waves after passing through the condenser.



The diode bridge is a bridge consisting of four diodes in the form of an electrical bridge, which unifies the output polarity regardless of the polarity of the input. The most common use of the diode bridge is to convert alternating current to direct current, that is, it acts as a rectifier. The bridge rectifier makes the output full wave rectified from the two AC input wires, resulting in a lower load compared to a rectifier that uses three wires for input. The main advantage of the diode bridge is that the output polarity is uniform regardless of the input polarity.

2N3906A	
1	Emitter
2	Base
3	Collector



موقع
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The transistor can act as an amplifier for small electrical signals or as a power amplifier. In both cases, the transistor amplifies an electrical signal that is delivered to its input circuit, and then transferred to its output circuit. It can also act as an electronic switch, the transistor shows a plateau-shaped frequency response to the frequencies of the input signal, as the frequency increases, the gain of the transistor rises to reach the highest possible value, then remains constant over a wide spectrum before falling back and decreasing again as the frequency increases above a specified threshold .



The shape of the waves after its exit from the transistor.



Conclusion.

This Everything what I can do.
Allah with you , good luck for you.

Reference .

I watched videos on youtube .

